

$$\begin{aligned}
& \triangleright f(b, c, r) := \left(\frac{1}{2} - \cos\left(\frac{5}{6}\pi - c - r\right) \cdot \sin(c) - \frac{1}{2} \cdot \cos\left(\frac{2 \cdot \pi}{3} + r\right) - \cos\left(\frac{\pi}{2} - b\right) \cdot \sin(b) \right)^2 \\
& \quad + \left(\sin\left(\frac{5}{6}\pi - c - r\right) \cdot \sin(c) - \left(\frac{1}{2} \cdot \sin\left(\frac{2 \cdot \pi}{3} + r\right) + \sin\left(\frac{\pi}{2} - b\right) \cdot \sin(b) \right) \right)^2 \\
& f := (b, c, r) \mapsto \left(\frac{1}{2} - \cos\left(\frac{5 \cdot \pi}{6} - c - r\right) \cdot \sin(c) - \frac{\cos\left(\frac{2 \cdot \pi}{3} + r\right)}{2} - \cos\left(-b + \frac{\pi}{2}\right) \right. \\
& \quad \left. \cdot \sin(b) \right)^2 + \left(\sin\left(\frac{5 \cdot \pi}{6} - c - r\right) \cdot \sin(c) - \frac{\sin\left(\frac{2 \cdot \pi}{3} + r\right)}{2} - \sin\left(-b + \frac{\pi}{2}\right) \cdot \sin(b) \right)^2
\end{aligned} \tag{1}$$

$$\begin{aligned}
& \triangleright g(b, c, r) := \text{simplify}(\text{expand}(f(b, c, 0)^2 - f(b, c, r) \cdot f(b, c, -r))) \\
& \quad g := (b, c, r) \mapsto \text{simplify}(\text{expand}(f(b, c, 0)^2 - f(b, c, r) \cdot f(b, c, -r)))
\end{aligned} \tag{2}$$

$$\begin{aligned}
& \triangleright \text{collect}(\text{expand}(g(b, c, r)), \cos(r)) \\
& - \frac{\cos(r)^2}{4} + \left(\sin(b) \cos(b) \sin(c) \cos(c) - \sin(b) \cos(b) \sqrt{3} \cos(c)^2 + \frac{\sqrt{3} \cos(b) \sin(b)}{2} \right. \\
& \quad - \sin(c) \cos(c) \sqrt{3} \cos(b)^2 - \cos(c)^2 \cos(b)^2 + \frac{\cos(b)^2}{2} + \frac{\sin(c) \sqrt{3} \cos(c)}{2} + \frac{\cos(c)^2}{2} \\
& \quad \left. - \frac{1}{4} \right) \cos(r) + \frac{\sin(b)^2}{2} - \sin(b) \cos(b) \sin(c) \cos(c) + \sin(b) \cos(b) \sqrt{3} \cos(c)^2 \\
& \quad - \frac{\sqrt{3} \cos(b) \sin(b)}{2} + \sin(c) \cos(c) \sqrt{3} \cos(b)^2 + \cos(c)^2 \cos(b)^2 - \frac{\sin(c) \sqrt{3} \cos(c)}{2} \\
& \quad - \frac{\cos(c)^2}{2}
\end{aligned} \tag{3}$$

$$\begin{aligned}
& \triangleright \text{simplify} \left(\frac{1}{r^2} \left(\left(-\frac{1}{4} \right) \cdot (1 - r^2) + \left(-\sqrt{3} \cos(c)^2 \cos(b) \sin(b) - \cos(b)^2 \cos(c)^2 \right. \right. \right. \\
& \quad + \frac{\cos(c)^2}{2} - \sqrt{3} \sin(c) \cos(c) \cos(b)^2 + \sin(c) \cos(c) \cos(b) \sin(b) \\
& \quad + \frac{\sin(c) \sqrt{3} \cos(c)}{2} + \frac{\sqrt{3} \sin(b) \cos(b)}{2} + \frac{\cos(b)^2}{2} - \frac{1}{4} \Big) \cdot \left(1 - \frac{r^2}{2} \right) \\
& \quad + \sqrt{3} \cos(c)^2 \cos(b) \sin(b) + \cos(b)^2 \cos(c)^2 + \sqrt{3} \sin(c) \cos(c) \cos(b)^2 \\
& \quad - \sin(c) \cos(c) \cos(b) \sin(b) - \frac{\sin(c) \sqrt{3} \cos(c)}{2} + \frac{\sin(c)^2}{2} - \frac{\sqrt{3} \sin(b) \cos(b)}{2} \\
& \quad \left. \left. - \frac{\cos(b)^2}{2} \right) \right) \\
& \frac{3}{8} + \frac{(2 \cos(b)^2 \sin(c) \cos(c) + \sin(b) (-1 + 2 \cos(c)^2) \cos(b) - \sin(c) \cos(c)) \sqrt{3}}{4}
\end{aligned} \tag{4}$$

$$+ \frac{(-1 + 2 \cos(c)^2) \cos(b)^2}{4} - \frac{\sin(b) \cos(b) \sin(c) \cos(c)}{2} - \frac{\cos(c)^2}{4}$$

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$$> h(b, c) := \frac{3}{8}$$

$$+ \frac{(2 \sin(c) \cos(c) \cos(b)^2 + \sin(b) (2 \cos(c)^2 - 1) \cos(b) - \sin(c) \cos(c)) \sqrt{3}}{4}$$

$$+ \frac{(2 \cos(c)^2 - 1) \cos(b)^2}{4} - \frac{\sin(c) \cos(c) \cos(b) \sin(b)}{2} - \frac{\cos(c)^2}{4}$$

$$h := (b, c) \mapsto \frac{3}{8} \quad (5)$$

$$+ \frac{(2 \cdot \sin(c) \cdot \cos(c) \cdot \cos(b)^2 + \sin(b) \cdot (2 \cdot \cos(c)^2 - 1) \cdot \cos(b) - \sin(c) \cdot \cos(c)) \cdot \sqrt{3}}{4}$$

$$+ \frac{(2 \cdot \cos(c)^2 - 1) \cdot \cos(b)^2}{4} - \frac{\sin(c) \cdot \cos(c) \cdot \cos(b) \cdot \sin(b)}{2} - \frac{\cos(c)^2}{4}$$

$$> \text{simplify} \left(\text{int} \left(\text{int} \left(\frac{h(b, c)}{\sin \left(b + c + \frac{\pi}{3} \right)^4}, b = 0 .. \frac{\pi}{6} \right), c = 0 .. \frac{\pi}{6} \right) \right) \\ - \frac{\ln(3)}{2} + \ln(2) \quad (6)$$

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$$> \text{int} \left(\text{int} \left(\frac{h(b, c)}{f(c, b, 0)^2}, b = 0 .. \frac{3.1415926535}{6}, \text{numeric} = \text{true} \right), c = 0 .. \frac{3.1415926535}{6}, \text{numeric} \right. \\ \left. = \text{true} \right)$$

$$0.1438410363 \quad (7)$$

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